Chapter 1: Foundations of Quantum Computing

Section 1.1: Introduction to Qubits

Quantum computing harnesses the phenomena of quantum mechanics to solve complex problems. A qubit is defined as the fundamental unit of quantum information, analogous to the classical bit. Unlike classical bits which can only be 0 or 1, a qubit can exist in a superposition of both states. Superposition allows quantum algorithms to process exponentially large state spaces simultaneously.

Quantum entanglement is defined as a physical phenomenon where pairs or groups of particles interact in ways such that the quantum state of each particle cannot be described independently.

```
import numpy as np
def hadamard_gate():
    return (1 / np.sqrt(2)) * np.array([[1, 1], [1, -1]])
state = hadamard_gate() @ np.array([1, 0])
print('Superposition state:', state)
```

Key Properties of Quantum Bits:
? State Superposition allows concurrent exploration of solutions.
? Quantum Entanglement links qubits regardless of physical distance.
? Quantum Interference amplifies correct probability amplitudes.



Chapter 2: Quantum Algorithms and Complexity

Section 2.1: Performance Comparison

Algorithm	Classical Complexity	Quantum Complexity
Shor's Algorithm	Exp(O(n^(1/3)))	Polynomial O(n^3)
Grover's Search	O(N)	O(sqrt(N))
Deutsch-Jozsa	O(2^(n-1))	O(1)

As demonstrated in the comparison table, quantum algorithms deliver quadratic to exponential speedups. Shor's algorithm poses a direct challenge to modern RSA cryptographic infrastructure.